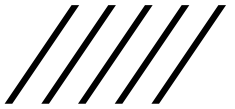
AI
SELECTED TOPICS

SECTION 1

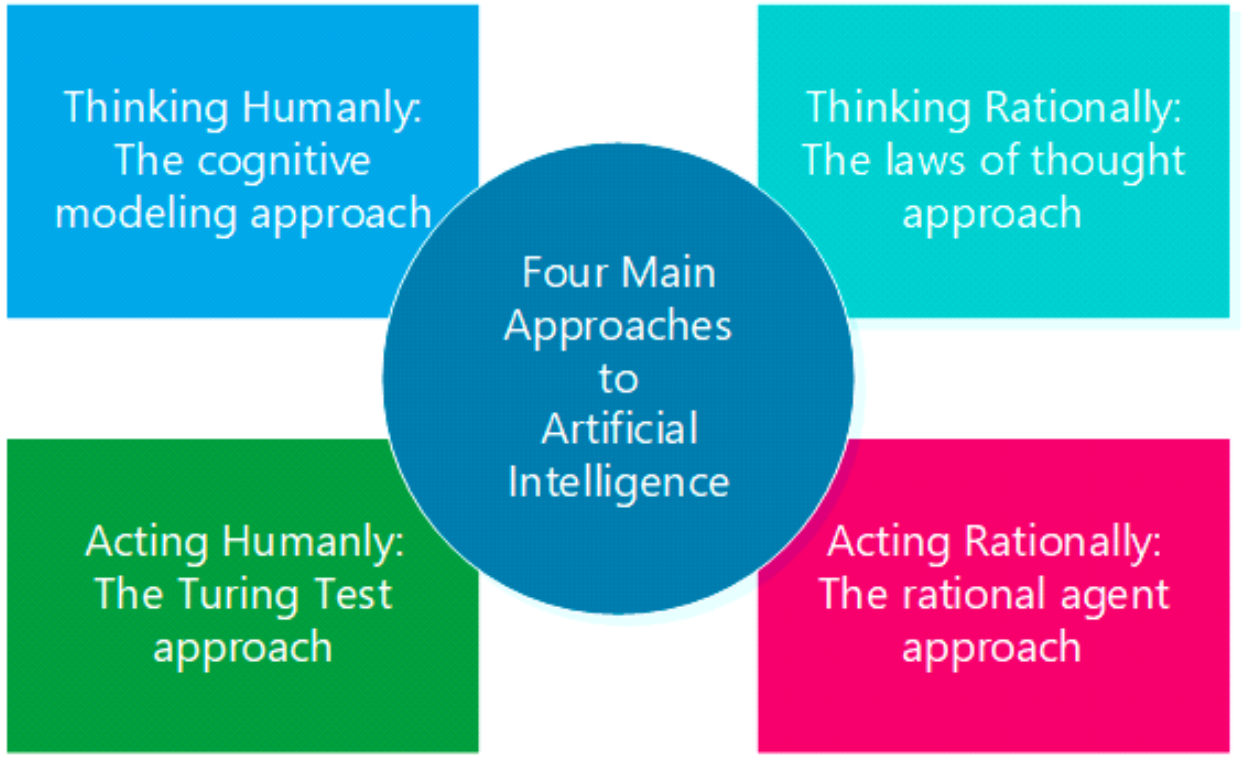
What is Intelligence?

- There are multiple definitions of intelligence:
  - The ability to learn and solve problems
  - The ability to think and act rationally
  - The ability to act like humans
- Some machines can appear to be intelligent from one perspective, but not intelligent from another perspective.



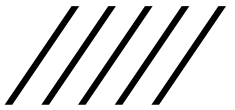
What is Artificial Intelligence?

- There are multiple definitions and approaches to AI.



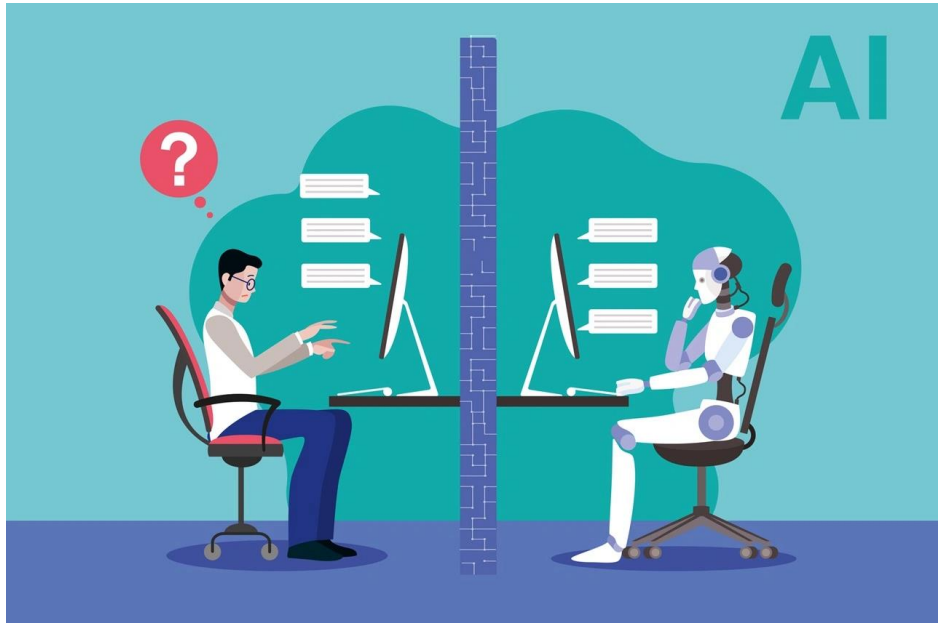
Thinking Humanely (Cognitive Modeling)

- AI is **“the automation of activities that we associate with human thinking, activities such as decision-making, problem solving, learning…”**
- Goal is to build systems that function internally in some way **similar to human mind**.



Acting Humanely (Turing Test)

- AI is “**The art of creating machines that perform functions that require intelligence when performed by people**”
- Alan Turing proposed the Turing Test in 1950 to determine whether a computer can be considered intelligent.



Acting Humanely (Turing Test)

- To pass the test, a computer would need to possess the following capabilities:
  - **Natural Language Processing:** to enable it to communicate successfully.
  - **Knowledge Representation:** to store what it knows or hears.
  - **Automated Reasoning:** to answer questions and draw conclusions.
  - **Machine Learning:** to detect and extrapolate patterns.
  - **Computer Vision:** to perceive objects.
  - **Robotics:** to manipulate objects and move about.

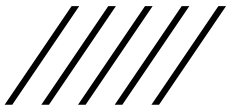
Total Turing Test

Rationality

- **Rationality** means **“doing the right thing”.**
- Take the **right/best action**, **achieve the best solution.**
- Depends on the quantity of **information** available.
- Sometimes, rational behavior requires **making assumptions**.

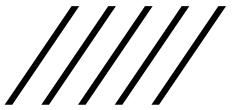
Thinking Rationally (Laws of Thought)

- AI is **“The study of the computations that make it possible to perceive, reasons, and act”**
- Think logically, rationally, and produce conclusions.



Acting Rationally (Rational Agent)

- AI is **“The branch of computer science that is concerned with the automation of intelligent behavior”**
- Rational agent acts to achieve **maximum value**.
- Emphasis is on **behavior**.



Fundamental Concerns of AI

- The two most fundamental concerns of AI researchers are **knowledge representation** and **search**.
- **Knowledge Representation:** the problem of capturing in a language the full range of knowledge required for intelligent behavior.
- **Search:** is a problem-solving technique that systematically explores a space of problem state.

Differences between Brain and Machine

Brain	Machine
1000 operations / sec	1,000,000 operations / sec
100,000,000,000 units	1-100 processors
Stochastic	Deterministic
Fault tolerant	Crashes
Evolves, learns	Designed, programmed

Important Features of AI

- Using computers to do **reasoning**, **pattern recognition**, **learning**, or some other form of **inference**.
- A focus on problems that **do not respond to algorithmic solutions**.
- A concern with problem-solving using **inexact, missing, or poorly defined information.**
- To reach answers that are not optimal but **“sufficient”** in some sense.

ANY QUESTIONS?

